[METRIC]
A-A-50800B
6 August 2019
SUPERSEDING
A-A-50800A
27 February 1998

COMMERCIAL ITEM DESCRIPTION

CADMIUM OXIDE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers one grade of cadmium oxide (CdO) for use in the preparation of cadmium plating solutions.

2. SALIENT CHARACTERISTICS

2.1 Assay. The product shall be at least 99 wt.% CdO with a minimum cadmium concentration of 87 wt.%. This assay shall be performed by an appropriate wet chemical (e.g., ethylenediaminetetraacetic acid titration) or instrumental (e.g., absorption/emission spectroscopy) method(s).

2.2 Trace metal impurities. The total concentration of silver, lead, tin, copper, mercury, zinc and iron in the product shall not exceed 0.050 wt.%. The total concentration of arsenic, antimony and thallium shall not exceed 0.005 wt.%. The impurity levels shall be determined using the appropriate instrumental analytical method(s) based on atomic or emission spectroscopy for solid materials (e.g., ark/spark atomic emission spectroscopy)

2.3 Volatile material. Approximately 20 grams (g) of the specimen shall be placed in a tared porcelain crucible and weighed to the nearest milligram (mg). The crucible and contents shall

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be heated in a muffle furnace at 105 ± 5 °C for two hours, cooled in a desiccator, and reweighed to the nearest mg. Volatile material shall be calculated as follows:

$$Volatile\ material\ (percent) = \frac{A - B}{A - T}\ X\ 100$$

Where: A = Weight of the crucible and contents before 105 °C conditioning (g)
B = Final weight of the crucible and contents (g)
T = Tare weight of the crucible (g)

The volatile material content of the cadmium oxide shall be not more than 0.50 percent by weight.

3. REGULATORY REQUIREMENTS

3.1 Safety data sheet (SDS). Manufacturers shall prepare and submit a SDS in accordance with FED-STD-313, "Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities", and meeting the requirements of 29 Code of Federal Regulations § 1910.1200, "Occupational Safety and Health Standards: Hazard Communication".

3.2 Marking, packaging, and labeling. The product shall be labeled, packed and marked in accordance with 49 CFR parts 100 to 185.

3.3 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.4 Toxic Substances Control Act (TSCA). The offeror shall have registered the product with the US Environmental Protection Agency (EPA) in accordance with 40 CFR parts 700-799, as applicable.

4. PRODUCT CONFORMANCE PROVISIONS

4.1 Product conformance. The product provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. A certificate of analysis (COA) shall be provided with each lot or batch of Product proving the salient characteristics of the product meets government requirements. The lot or batch number and the date of inspection/analysis from the offer shall be on each COA.

4.2 Market acceptability. The product offered must have been sold to the Government or commercial market.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

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6. NOTES

(This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Source of documents.

6.1.1 Electronic copies of CFR documents can be obtained from https://gov.ecfr.io/cgi-bin/ECFR.

6.1.2 Electronic copies of FAR documents may be obtained from https://www.acquisition.gov/browse/index/far.

6.1.3 Copies of FED-STD-313 can be obtained from https://quicksearch.dla.mil.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, name, and date of this specification.
b. Unit and quantity required.
c. Packaging requirements.

6.3 National stock number (NSN). NSN 6810-00-281-8804 corresponds to this CID. This may not be the only NSN associated with this document.

6.4 Key Words.

Electroplating
Plating
Corrosion resistance

MILITARY INTERESTS:

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - 10FTE

Custodians:
Navy – AS
Air Force – 68
DLA – GS

Preparing Activity:
DLA – GS3

(Project 6810-2019-003)

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